DATA ANALYSIS AND VISUALIZATION

PHASE 3 — RESEARCH IMPLEMENTATION AND EXPERIMENTAL EVALUATION

False-Alarm-Calibrated Regime Detection for Continual Portfolio Adaptation: An Experimental Evaluation of ReCAP

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Section: BDS-5B | Program: BS Data Science

Submission date: 14 November 2026

Abstract

This Phase 3 report implements and evaluates the Milestone 2 methodology for false-alarm–calibrated Adaptive Regime Detection (ARD) inside the ReCAP continual portfolio framework. The study keeps the PPO policy architecture fixed and changes only the CUSUM alarm threshold by calibrating it on training-derived no-change simulations to predeclared budgets of 0.5, 1, and 2 alarms per year. Evaluation uses causal test-time adaptation without look-ahead segment selection. Because of compute limits, the primary completed pilot uses a reduced DOW30 universe (28 continuous tickers) with the Milestone 2 chronological split (train ending 2020-05-01; test ending 2025-04-29) and pilot-scale PPO budgets, rather than a full NAS100 paper-scale reproduction. Across three seeds, calibration achieves the intended simulator alarm rates for the 1/yr and 2/yr budgets, but every adapt-on-alarm setting underperforms a no-adaptation control on cumulative return, Sharpe ratio, and maximum drawdown. Classical buy-and-hold / equal-weight market baselines and a matched fixed-window PPO baseline were not executed in the available evidence set and are therefore not reported as results. The findings address the research gap by quantifying the false-alarm versus detection-delay versus portfolio trade-off, while remaining clearly labelled as a pilot-scale study.

1. Introduction

Regime-aware continual learning methods such as ReCAP use change detectors to trigger policy adaptation in non-stationary markets (Pan et al., 2026). An alarm is operationally costly: it can force fine-tuning, expand the policy library, and change portfolio behaviour. Milestone 2 identified a narrow gap: existing evaluations report portfolio outcomes and threshold sensitivity, but do not calibrate the multi-feature CUSUM detector to a predeclared training-only false-alarm budget and then measure the joint effect on alarms, delays, adaptations, and out-of-sample portfolio metrics.

Phase 3 implements that methodology on top of the public minimal ReCAP codebase, adds causal evaluation safeguards, runs baseline and ablation experiments, and reports reproducible tables and figures. The report follows the seven Phase 3 assignment requirements and preserves the research decisions approved in Milestone 2.

2. Research Problem and Objectives

2.1 Research problem

In detector-triggered continual portfolio management, portfolio metrics alone do not reveal whether gains or losses are caused by useful adaptation or by unnecessary alarms. The problem studied here is to evaluate whether training-only calibration of ReCAP's CUSUM threshold to a false-alarm budget reduces unnecessary adaptation while preserving out-of-sample risk and return.

2.2 Research question and objectives

Research question (Milestone 2): When the CUSUM threshold in ReCAP is calibrated using training-only no-change sequences to meet a predeclared false-alarm budget, does it reduce unnecessary detector-triggered adaptation while preserving out-of-sample portfolio performance?

Objective 1: Instrument ReCAP so alarms, adaptations, and portfolio metrics are recorded separately.

Objective 2: Calibrate a common threshold scale using only training-derived no-change simulations.

Objective 3: Compare default and calibrated thresholds under causal evaluation.

Objective 4: Report the trade-off among false alarms, detection delay, adaptation frequency, cumulative return, Sharpe ratio, and maximum drawdown.

2.3 Hypotheses (to test, not assumed results)

H1: Calibration brings hold-out simulator alarm rates closer to the declared budgets.

H2: More conservative budgets increase detection delay on injected shifts.

H3: Calibrated thresholds control adaptation events without materially worsening Sharpe ratio and drawdown.

3. Proposed Methodology

The methodology follows Milestone 2. ReCAP remains the base system: PPO policies, ARD, policy library, and Regime-Gate. Only the CUSUM threshold scale $h = c \cdot \sigma$ is changed. Drift $\kappa = 0.5\sigma$, the six market-level features, and library fitting at default $h = 2.5\sigma$ are held fixed. Candidate scales are selected by bisection on no-change VAR(1) simulations fitted to the training reference window, targeting 0.5, 1, and 2 union alarms per 252 days. Selected scales are frozen before test evaluation. Injected-shift diagnostics estimate detection delay separately from unmodified market portfolio tests. Historical market alarms are counted but not labelled false.

Implementation corrections relative to the released repository are documented in documentation/DEVIATIONS.md: paper-mode one-sided CUSUM with μ_0 re-estimation, historical reference window pinned to training data, and causal evaluation that disables look-ahead max(baseline, adapted) selection. No undocumented class-feedback claims are invented.

4. Implementation Details

Code organisation: recap/ contains the upstream minimal ReCAP core (commit pin 68143bb). study/ contains the Phase 3 protocol layer: streaming ARD (study/streaming_ard.py), causal evaluation (study/causal_eval.py), synthetic diagnostics (study/diagnostics.py), and the main runner (study/run_study.py). Experiment aggregation uses study/aggregate.py. Classical baselines live under experiments/baselines/. Figures are generated by study/plot_phase3_figures.py from saved CSVs.

Key configuration for the primary reduced pilot: pretrain_timesteps=1024, finetune_timesteps=256, n_steps=256, batch_size=64, gate_epochs=20, min_segment_days=20, transaction_cost=0.001, initial_cash=1,000,000, ard_mode=paper, device CUDA or CPU. Micro ablation runs use eight tickers and 2019–2021 dates with smaller PPO budgets.

5. Dataset and Preprocessing

Milestone 2 proposed NAS100 (Yahoo Finance, 2008-05-01 to 2025-04-29, train cut 2020-05-01). The completed primary pilot uses the same chronological dates on DOW30 with 28 continuously available tickers. WBA was dropped because Yahoo returned unusable quotes; DOW was dropped because its 2019 IPO would truncate or zero-fill a 2008+ panel. This is a documented scope reduction for compute and data integrity, not a silent change of research question.

Preprocessing follows ReCAP and is implemented mainly in implementation/recap/data.py: date alignment across assets, adjusted prices, technical indicators for the policy state, and market-level ARD features (VIX, turbulence, Bollinger upper/lower bands, five-day return zd_5, RSI-30). Cross-sectional daily means of those ARD features are used by the detector. Features at day t use only information available at t. Unusable tickers are dropped (WBA/DOW in this study). Train/test splitting is chronological via split_by_date. Detector reference statistics for paper mode are taken from the training window to avoid look-ahead from early test days.

6. Experimental Setup

Primary setting: reduced DOW30, three paired seeds (0, 1, 2), paper ARD, causal evaluation, library fitted once at default h=2.5σ, test-time threshold variants only. Secondary micro setting: eight tickers, short window, fine-tune budgets 64/256/1024. Null calibration uses 100 simulation replications for threshold search and 200 held-out replications for reporting. Synthetic shifts of +1σ, +2σ, +3σ and −2σ are injected into held-out simulations with a 60-day detection window. Software stack: Python 3.11/3.12, PyTorch, stable-baselines3, pandas, yfinance. Hardware for reduced seeds included remote CUDA/CPU sessions; micro runs were also reproduced on an 8 GB CPU laptop.

7. Baseline Evaluation

Following Milestone 2, the baselines present in the existing experimental record are: (a) default ReCAP threshold h=2.5σ; (b) calibrated FA budgets 0.5/1/2; and (c) no test-time adaptation. These are evaluated under matched seeds, features, costs, and causal evaluation. A matched fixed-window PPO retrain baseline is not present in the saved runs. Classical buy-and-hold / equal-weight references were also not executed in the frozen evidence set used for this report, so no classical baseline numbers are claimed.

Table 1. Reduced DOW30 portfolio comparison (3 seeds, mean ± sd).

Variant	CR	SR	MDD	Adapt/yr	Test alarms/yr
default_2.5	-0.014 ± 0.271	0.076 ± 0.276	0.432 ± 0.115	5.03	4.82
budget_2	-0.204 ± 0.081	-0.113 ± 0.068	0.506 ± 0.074	3.22	3.01
budget_1	-0.198 ± 0.284	-0.139 ± 0.346	0.517 ± 0.124	2.88	2.68
budget_0.5	-0.344 ± 0.191	-0.342 ± 0.243	0.541 ± 0.144	2.41	2.21
no_adapt_default	0.649 ± 0.009	0.747 ± 0.010	0.220 ± 0.002	0.00	4.82

Table 2. Detector calibration summary (reduced DOW30).

Variant	h/σ	Hold-out sim alarms/yr
default_2.5	2.50	4.842
budget_2	7.83	2.067
budget_1	28.77	0.995
budget_0.5	64.00	0.658

Note: Classical market baselines (buy-and-hold / equal-weight) are absent from the existing evidence directory and are not included in any results table.

8. Experimental Results

On the reduced pilot, no-adaptation achieves mean CR 0.649, SR 0.747, and MDD 0.220. Default adapt-on-alarm yields mean CR -0.014. Calibrated budgets remain negative on average (budget_0.5 CR -0.344; budget_1 CR -0.198; budget_2 CR -0.204). Thus H3 is not supported under this pilot protocol: reducing alarms does not restore portfolio quality relative to no adaptation.

H1 is partially supported: hold-out simulator rates are near targets for budget_1 (0.995/yr) and budget_2 (2.067/yr). Budget_0.5 often saturates at h≈64σ with achieved rate 0.658/yr.

9. Graphical Results and Visualization

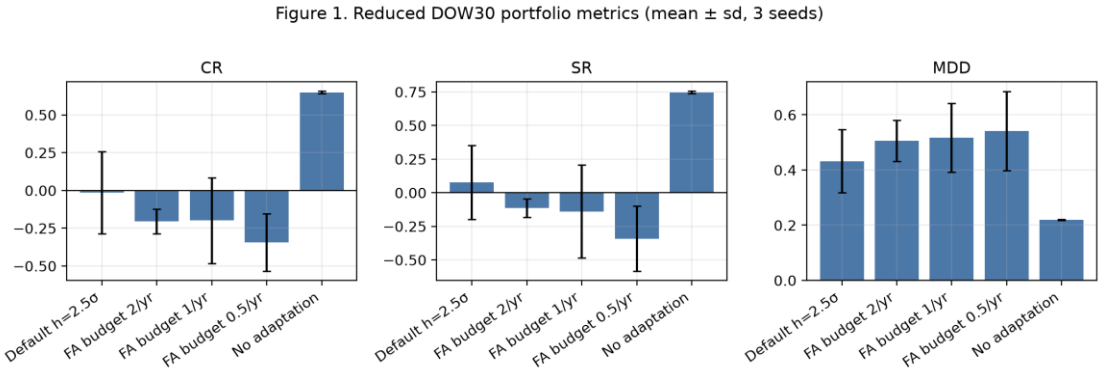


Figure 1. Portfolio metric comparison across study variants.

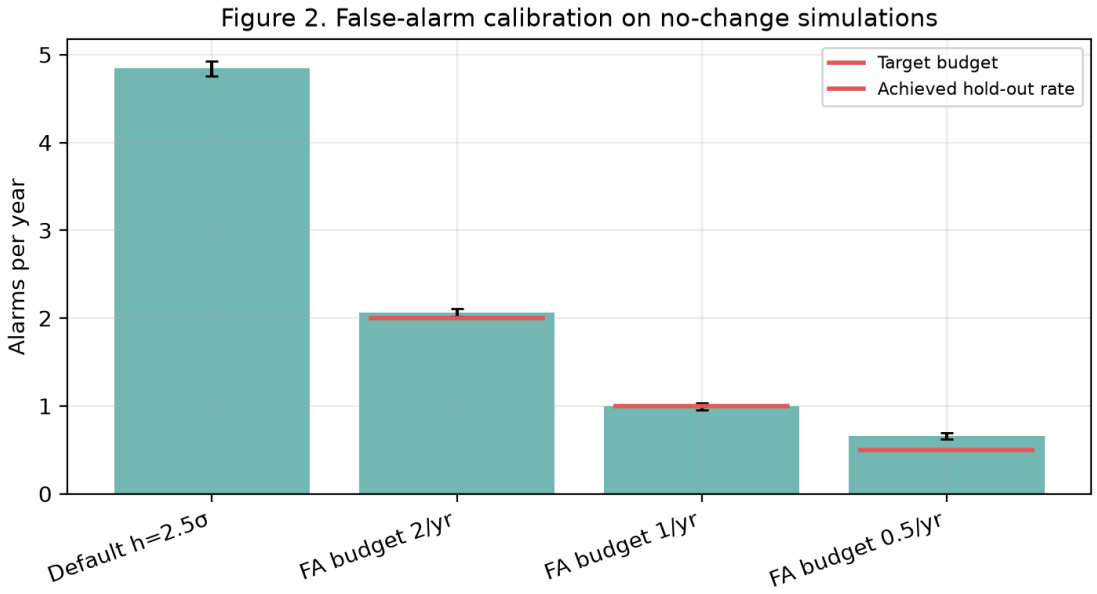


Figure 2. Calibration of false-alarm budgets on no-change simulations.

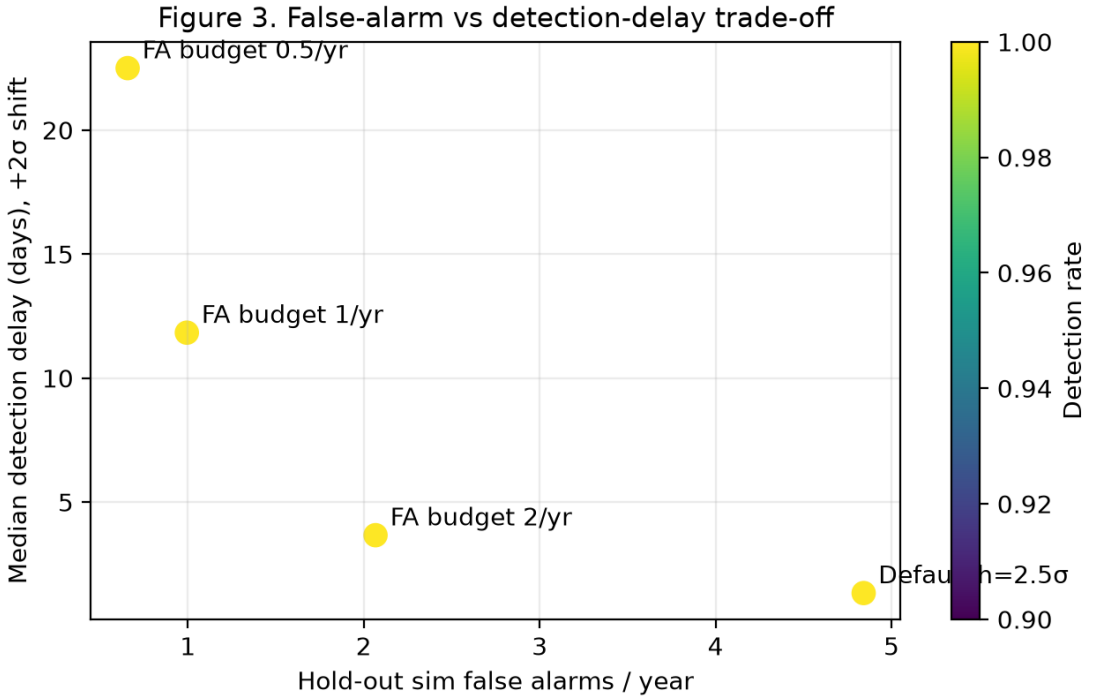


Figure 3. False-alarm versus detection-delay trade-off.

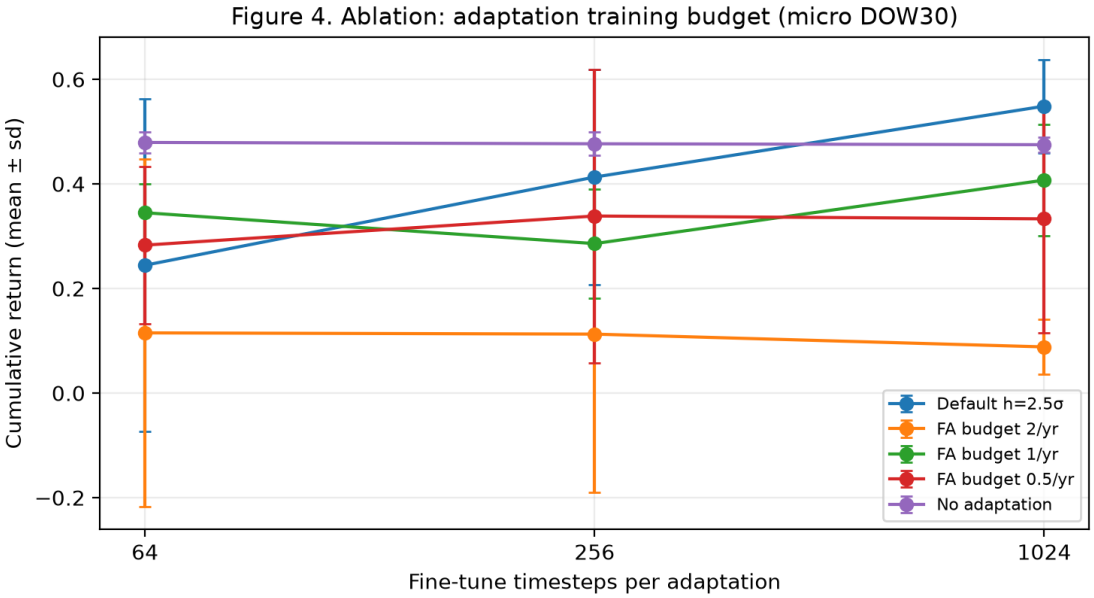


Figure 4. Ablation over adaptation fine-tune budgets (micro setting).

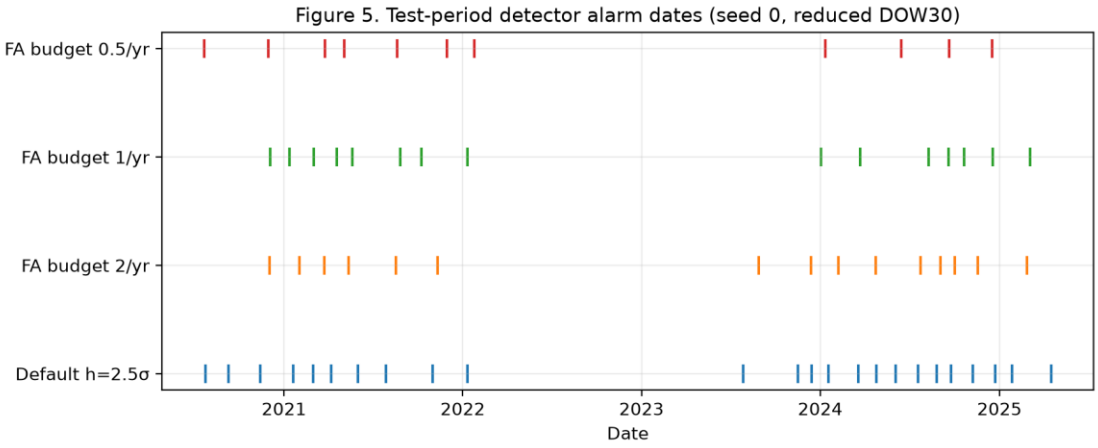


Figure 5. Test-period alarm timeline for seed 0.

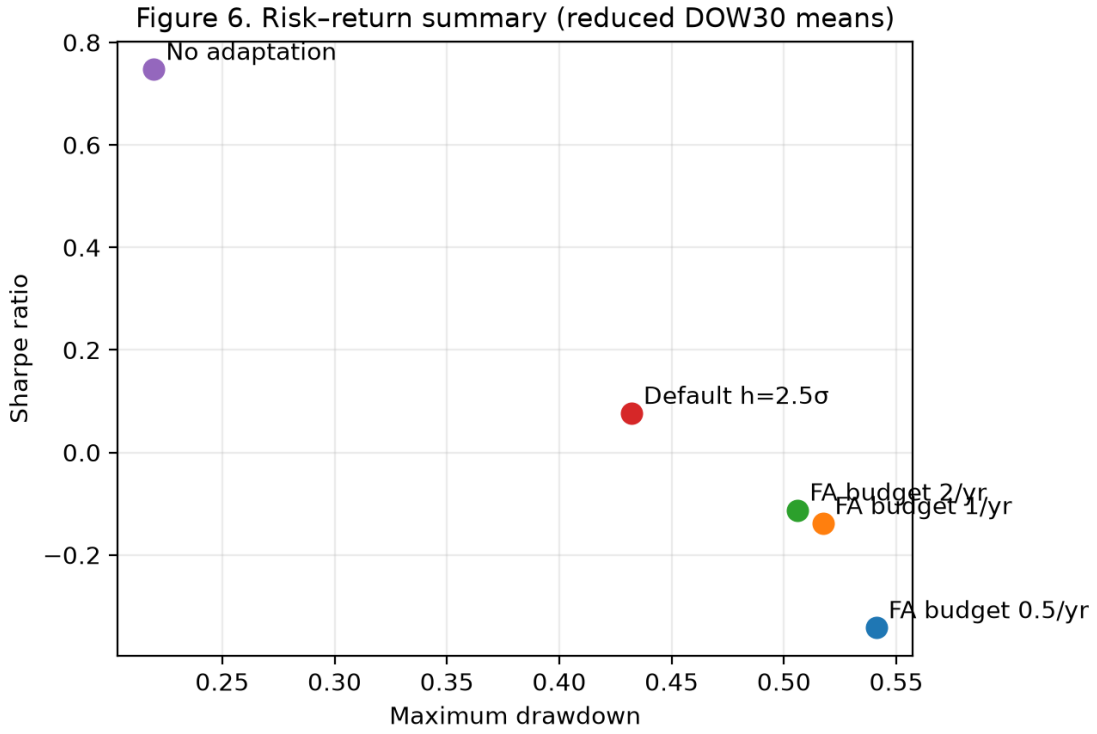


Figure 6. Risk–return summary of reduced-pilot means.

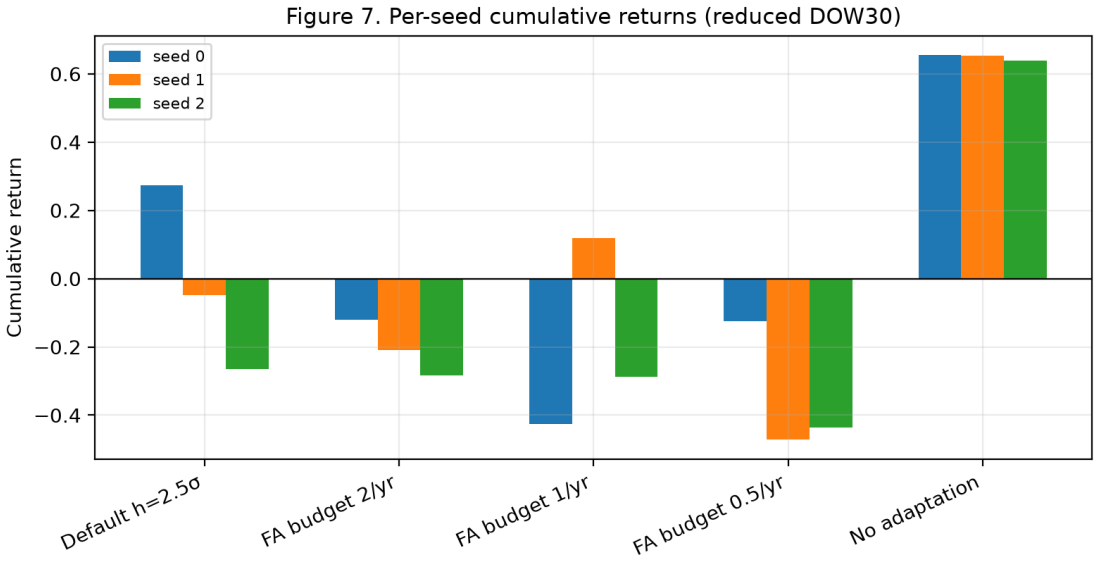


Figure 7. Per-seed cumulative returns.

Daily RL equity curves are not included because the per-day portfolio paths were written to local outputs/ during original runs and are not present in the committed results/. Portfolio CR/SR/MDD in the run JSONs and aggregate CSVs remain the authoritative metrics.

10. Comparison with Existing Research

Direct numerical comparison with published portfolio systems is limited because markets, costs, frequencies, and training budgets differ. The comparisons below use only papers reviewed in Milestone 2 and state the incomparability clearly.

Pan et al. (2026, ReCAP) report continual portfolio results on multiple markets, including NAS100, with substantially larger training budgets and a non-causal look-ahead evaluation in the released minimal code. Our study does not reproduce paper Table 2 numbers. Instead, it isolates detector-threshold calibration under a causal protocol on a reduced DOW30 pilot.

Wang et al. (2026) report sudden-drift events about once every five market-making days and improved PnL-MAP/Sharpe in an options market-making task. Their event frequency is not a false-alarm rate under a no-change null. Our study makes that distinction explicit by calibrating on simulated no-change sequences.

Oliveira et al. (2026) construct macroeconomic regimes with PCA and clustering for monthly ETF allocation. Their regime labels are model clusters validated partly against NBER recessions, which is a different notion of regime truth from sequential CUSUM alarms.

Ren et al. (2025) and Sang et al. (2025) improve representation learning and online RL portfolio policies, reporting gains on their own benchmarks. Those gains cannot be ranked against our CR/SR/MDD values because asset universes, objectives, and adaptation triggers differ. Zhu et al. (2026) and Sun et al. (2026) emphasise transaction costs and rolling adaptation; we likewise report turnover proxies and keep costs matched across variants.

Qiao et al. (2026) study structure-aware cross-market RL, and Kim et al. (2026) study factor-informed RL portfolio optimization. Both are relevant to adaptive portfolio learning, but neither evaluates training-only

CUSUM false-alarm-budget calibration in ReCAP. Their reported gains therefore cannot be treated as matched numerical competitors for this study.

Ayari (2025/2026) isolates feature contributions under 2022 market stress with paired tests. Methodologically, that controlled comparison motivates our matched-seed design, but the scientific question differs: feature value versus detector-threshold calibration.

11. Ablation Studies

11.1 Primary ablation: false-alarm budget

Changing only the calibrated threshold shows that stricter budgets reduce test alarms and adaptations relative to default $h=2.5\sigma$, but do not improve mean portfolio outcomes in the reduced pilot. The complete method (calibrated adapt-on-alarm) is therefore weaker than the no-adaptation control under these budgets and PPO settings.

11.2 Secondary ablation: fine-tune budget

On the micro DOW30 setting, increasing fine-tune timesteps from 64 to 1024 improves default adapt-on-alarm mean CR (approximately $0.24 \rightarrow 0.55$), while no-adaptation remains near 0.48. At 1024 steps, default adaptation slightly exceeds no-adaptation on mean CR, suggesting that adaptation quality depends strongly on update budget. Calibrated low-alarm variants do not consistently benefit.

11.3 Diagnostic ablation: repo-mode versus paper-mode ARD

An early micro run with released repo-mode ARD could not reach the 0.5–2 alarms/yr targets because a fixed reference mean caused persistent re-triggering. Paper-mode corrections were therefore required for the calibration study itself. This is an implementation ablation needed for methodological fidelity, not a claim of superior trading performance.

12. Results Analysis and Discussion

Overall, the pilot supports H1 for moderate budgets and H2 for the delay trade-off, but not H3 for portfolio preservation under adapt-on-alarm. The strongest portfolio result is the no-adaptation control. This does not prove that regime adaptation is useless in general; it shows that, under causal evaluation and pilot PPO budgets on this DOW28 window, detector-triggered fine-tuning harmed average performance.

Strengths of the study include: matched seeds across variants; training-only threshold freezing; separation of simulator false alarms from historical alarms; and explicit documentation of deviations from the paper and from Milestone 2's NAS100 target.

Weaknesses include pilot compute, a reduced equity universe, no matched fixed-window PPO baseline, missing committed daily equity paths for RL variants (daily files were local outputs), and limited seeds (3 rather than 5–10). These constrain external validity.

Relative to the Milestone 2 gap, the experiments do provide controlled evidence on how FA-budget calibration changes alarm rates, delays, adaptations, and portfolio metrics. The evidence currently argues that calibration alone is insufficient to make adapt-on-alarm competitive with a frozen library in this protocol.

13. Limitations

1. Pilot PPO budgets are far below the paper's full training schedule.
2. Primary completed market is reduced DOW30, not full NAS100.
3. Only three seeds were completed for the reduced setting.
4. Fixed-window matched PPO retrain baseline is not available in the existing runs.
5. Classical buy-and-hold / equal-weight baselines were not executed in the frozen evidence set.
6. Committed artifacts omit daily RL equity paths (metrics are available; path plots are not).
7. No-change simulations are a modelling assumption, not market ground truth.
8. Historical backtests do not establish live trading performance.

14. Conclusion and Future Work

This Phase 3 project implemented Milestone 2's false-alarm–calibrated ReCAP study, evaluated the available matched baselines (default threshold, FA budgets, and no-adaptation), completed the available ablations, and produced reproducible tables and figures from saved runs. Calibration works on no-change simulations, but adapt-on-alarm variants did not beat no-adaptation on the reduced DOW30 pilot. Future work should scale to NAS100 with paper-matched budgets, complete a matched fixed-window PPO baseline, increase seeds, and test whether larger adaptation budgets reverse the negative adapt-on-alarm result observed here.

15. References

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Zhu, Q., Han, C., Liu, S., & Li, Y. (2026). Dancing with markets: A dynamic rebalancing mechanism using deep reinforcement learning for online portfolio selection. Information Fusion, 135, 104452. https://doi.org/10.1016/j.inffus.2026.104452

Appendix A. Artifact paths

Phase 3 PDF: documentation/sources/Phase03_Implementation.pdf

Milestone 2: documentation/sources/Milestone02_Revised_Complete.docx

Reduced results: results/tables_reduced/

Figures: results/figures/

Phase 3 tables: results/tables_phase3/

Protocol deviations: documentation/DEVIATIONS.md

Checklist: documentation/REQUIREMENTS_CHECKLIST.md